

Maximum Drawdown: Measuring Downside Risk

Maximum Drawdown (MDD) measures the largest peak-to-trough decline in your strategy's value. It answers the critical question:

"What's the worst decline my strategy has experienced?"

Think of Maximum Drawdown as the steepest drop on a roller coaster. Even if the overall ride is enjoyable, a drop that's too steep might be more than some passengers can handle. Similarly, even a profitable strategy might have drawdowns that exceed your risk tolerance.

```
1. def calculate_max_drawdown(returns):
2.     # Calculate cumulative returns
3.     cumulative_returns = (1 + returns).cumprod()
4.
5.     # Calculate running maximum
6.     running_max = cumulative_returns.cummax()
7.
8.     # Calculate drawdown
9.     drawdown = (cumulative_returns / running_max) - 1
10.
11.    # Get maximum drawdown
12.    max_drawdown = drawdown.min()
13.
14.    return max_drawdown
15.
16. # Calculate maximum drawdown
17. max_drawdown = calculate_max_drawdown(strategy_returns)
18. print(f"Maximum Drawdown: {max_drawdown:.2%}")
19.
```

Maximum Drawdown is typically expressed as a percentage. A -20% MDD means that at its worst point, the strategy lost 20% from its previous peak. Lower (less negative) MDDs are generally better, but they must be viewed in the context of returns—higher returns often come with larger drawdowns.